

Lectures	Mondays & Wednesdays, 5:45 - 7:45 pm, UWBB-230
Instructor	Jim Hogg
Office Hours	Wednesdays, 8:00-9:00 pm, UWBB-107G (or by appointment)
Email	jimhogg@uw.edu
Phone	425 590 7815

Course Description

This course introduces Compilers and Interpreters: what they are, and how they work. We cover both theory and practice for: Lexing, Parsing, ASTs, Interpretation, Codegen and Optimization. The class project builds a Compiler for a subset of the C programming language, generating ARM-32 machine code.

Grading

Homework: 30%, Projects: 40%, Midterm: 15%, Final: 15%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide.

Textbook

The lectures will cover all of the material needed to understand the course. However, students may find it helpful to refer to a text book. There are dozens of good books available on compilers. For example:

[Engineering a Compiler](#), by Cooper & Torczon, Second Edition, 2011. This is solid and understandable, with lots of practical advice. It covers much more than needed for CSS-448.

Due Dates & Times

All Homework and Projects should be submitted to Canvas by midnight of the date given on the class calendar (always a Friday, so easy to remember)

Policies

Attendance: Attend all classes.

Assignment Submission: Strict deadlines. Follow submission instructions. If you need an extension due to exceptional circumstances, email or talk to me. If your coding assignment does not work as submitted, you will receive no marks for that

assignment, so always download the assignment back from Canvas in order to check the upload did not lose anything!

Effort: Expect to spend 8-12 hours per week outside class.

Academic Integrity: Do the right thing. If in doubt, see <http://guides.lib.uw.edu/bothell/ai> for details.

Communication: Send me email, or open a Discussion on Canvas, for clarification on any questions. If the answer to your question could benefit the rest of the class, I will post my response to Canvas. For anything more complicated, please come along to Office Hours.

Problems: If you are having difficulties, talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Access and Accommodations: Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course. See <http://www.uwb.edu/studentaffairs/drs> if you need to establish accommodations.

Common Course Policies for the School of STEM: See <https://www.uwb.edu/stem/about/stem-policies> for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, For Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity, Student Support Services, Wondering How to Address Faculty? etc.

Course Material: All lectures, home works and projects will become available, as the weeks go by, on this website, under **Files**.